

POWER BI Logistics & Supply Chain Analytics Project

SQL Data Import Link:

https://file.notion.so/f/f9fda094-a3c7-4f29-953c-59c615317d5b/0aaa7d69-60ab-416f-8421-d7bb3ca52417/logistics_supply_chain_inventory_management.sql?table=block&id=35a62e44-fc3f-805b-ab48-c89ab71eb55e&spaceId=f9fda094-a3c7-4f29-953c-59c615317d5b&expirationTimestamp=1778508000000&signature=pS9p2jhpHdUXs0nxjot24Hu2_hHhUhHWkMb28vTu2n4&downloadName=logistics_supply_chain_inventory_management.sql

Business Scenario

A US-based logistics organization is facing multiple warehouse and inventory management challenges across different distribution centers.

The company wants to build a complete Power BI reporting solution to monitor:

Inventory movement

Stock availability

Warehouse efficiency

Inventory discrepancies

Validation issues

Task productivity

Supplier performance

Operational KPIs

The organization has shared PostgreSQL database data from multiple warehouse systems and expects the Power BI team to create interactive dashboards for operations teams, warehouse managers, quality teams, and executive leadership.

Objective of the Project

Students need to:

Connect PostgreSQL database with Power BI

Build proper data model relationships

Create analytical dashboards

Write DAX measures

Identify operational problems

Generate actionable business insights

Build executive-level reporting dashboards

Dataset Overview

Available Tables

Table Name	Purpose
item_categories	Product category master
suppliers	Supplier information
warehouses	Warehouse master
warehouse_staff	Staff information
inventory_item_master	Inventory item details
inventory_movement_log	Stock movement transactions
inventory_profiling	Inventory stock snapshots
inventory_discrepancy_log	Stock mismatch tracking
inventory_validation_log	Inventory validation issues
task_allocation_log	Warehouse task tracking
inventory_kpi_summary	Monthly KPI summary

Available Power BI Views

View Name	Purpose
vw_item_master_detail	Item-level detailed view
vw_inventory_movement_summary	Movement analytics
vw_inventory_profile	Inventory health analysis
vw_discrepancy_analysis	Discrepancy reporting
vw_validation_issue_summary	Validation issue analysis
vw_task_productivity_summary	Staff productivity
vw_inventory_kpi_summary	Executive KPI dashboard

Corporate Requirement Mail – 1



From: Warehouse Operations Team



Organization: NovaChain Logistics Inc., USA



Subject: Requirement for Basic Inventory Overview Dashboard in Power BI



Dear Power BI Development Team,

We are currently facing challenges in monitoring our warehouse inventory operations across multiple distribution centers in the US.

Our operations team requires a **basic inventory overview dashboard** to monitor stock distribution, inventory categories, warehouse utilization, and supplier contribution.

At this stage, we only require a **basic analytical dashboard** without advanced DAX calculations. The goal is to create a centralized operational reporting dashboard for warehouse managers.



Business Objective

Build a Power BI dashboard that helps warehouse operations teams understand:

- Overall inventory health
- Warehouse stock distribution
- Item category contribution
- Supplier contribution
- Inventory availability
- Stock status overview

The dashboard should be simple, interactive, and management-friendly.



Tables / Views Required

Tables

- inventory_item_master
- inventory_profiling
- warehouses
- suppliers
- item_categories

Views

- vw_item_master_detail
 - vw_inventory_profile
-

1 Inventory Overview

What is the total inventory stock value currently available across all warehouses?

2 Item Status Analysis

How many inventory items are:

- Active
 - Inactive
 - Discontinued
-

3 Warehouse Performance

Which warehouse currently holds the highest inventory stock value?

4 Category Distribution

Which inventory categories contain the maximum number of items?

5 Warehouse-wise Inventory Distribution

Show warehouse-wise distribution of:

- Total stock units
 - Stock value
 - Number of inventory items
-

6 Supplier Contribution Analysis

Which suppliers are contributing the highest number of inventory items?

7 Inventory Health Analysis

Analyze stock status distribution:

- Healthy
 - Overstock
 - Stockout
 - Slow Moving
 - Below Reorder
-

Category vs Warehouse Matrix

Create a matrix showing:

- Warehouse
- Category
- Total Stock Value
- Closing Stock

Mail 2 – Inventory Movement & Stock Analytics

 Dear Power BI Development Team,

After reviewing the initial inventory overview dashboard, management now wants deeper analytical insights into inventory movement, warehouse stock behavior, stock availability, and inventory turnover patterns across all warehouses.

The operations leadership team has identified several operational concerns related to:

- Frequent stockouts
- Slow-moving inventory
- Overstock conditions
- Uneven warehouse movement
- Inventory flow inefficiencies

For this phase, the dashboard should focus heavily on:

- Proper Data Modeling
- DAX Measures
- Time-based analysis
- Warehouse movement analytics
- Inventory performance tracking

The goal is to help warehouse managers and operations analysts identify inventory movement trends and operational bottlenecks using Power BI.

Tables / Views Required

Tables

- inventory_movement_log
- inventory_profiling
- inventory_kpi_summary
- warehouses
- item_categories
- inventory_item_master

Views

- vw_inventory_movement_summary
- vw_inventory_profile
- vw_inventory_kpi_summary

1 Monthly Inventory Movement Trend

Analyze monthly stock movement trends across all warehouses.

Identify:

- High movement months
 - Low movement months
 - Sudden movement spikes
-

2 Warehouse Movement Performance

Which warehouses have the highest inventory movement volume?

Compare:

- Total movement quantity
 - Total movement value
 - Movement frequency
-

3 Movement Type Analysis

Which movement type occurs most frequently?

- Stock In
 - Stock Out
 - Transfer
 - Purchase Receipt
 - Sales Issue
 - Return In
 - Damage Write-off
 - Adjustment
-

4 Slow-Moving Inventory Analysis

Identify inventory items and categories with:

- High stock levels
 - Low stock-out movement
 - Slow inventory turnover
-

5 Stockout Analysis

Which warehouses are facing the highest stockout conditions?

Analyze:

- Stockout frequency
 - Warehouse impact
 - Category impact
-

6 Inventory Turnover KPI

Create DAX measures to calculate:

- Inventory turnover ratio
- Average stock movement
- Monthly stock flow

Identify low-performing categories.

7 Inventory Value Movement Analysis

Which categories generate the highest inventory value movement?

Compare:

- Stock value moved
 - Quantity moved
 - Warehouse-wise contribution
-

8 Overstock vs Healthy Inventory

Analyze warehouse inventory profile and identify:

- Overstock items
- Healthy inventory
- Below reorder inventory

Create a comparison dashboard.

Mail 3 – Discrepancy, Validation & Task Productivity

Analytics

👋 Dear Power BI Development Team,

After reviewing inventory movement and stock analytics dashboards, our warehouse quality and operations leadership team now wants deeper operational analysis related to inventory discrepancies, validation failures, warehouse process quality, and staff productivity.

Recently, several warehouses have reported:

- High inventory mismatches
- Validation failures
- Barcode-related issues
- Delayed warehouse tasks
- Increased operational inefficiencies

Management wants a detailed analytical dashboard that can help identify:

- Root causes of discrepancies
- Validation issue patterns
- Warehouse operational bottlenecks
- Staff productivity performance
- Delayed task behavior

For this phase, the dashboard must focus heavily on:

- Intermediate DAX
- Root Cause Analysis
- Ranking Logic
- Drill-through Reporting
- Operational KPI Analysis
- Data Modeling Optimization

The goal is to help warehouse quality teams and operations managers reduce inventory inconsistencies and improve operational efficiency using Power BI.

Tables / Views Required

Tables

- inventory_discrepancy_log
- inventory_validation_log
- task_allocation_log
- warehouse_staff
- warehouses
- inventory_item_master
- item_categories

Views

- vw_discrepancy_analysis
- vw_validation_issue_summary
- vw_task_productivity_summary

Business Questions / Tasks

Warehouse Discrepancy Analysis

Which warehouses have the highest discrepancy value?

Analyze:

- Total discrepancy quantity
 - Total discrepancy value
 - Severity distribution
-

Top Discrepancy Reasons

Identify the most common discrepancy reasons:

- Cycle Count Error
- Picking Error
- Barcode Scan Error
- Supplier Shortage
- Wrong Bin Location
- Damaged Stock

Rank the top operational issues.

Category-wise Inventory Mismatch

Which inventory categories have the highest mismatch frequency?

Compare:

- Discrepancy quantity
 - Discrepancy value
 - Severity levels
-

4 Validation Issue Analysis

Which warehouses are generating the highest validation issues?

Analyze:

- Open issues
 - Closed issues
 - High severity issues
 - Validation trends by month
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5 Validation Type Root Cause Analysis

Which validation issue types occur most frequently?

- Invalid Barcode
- Negative Stock
- Missing Supplier Mapping
- Duplicate SKU
- Location Mismatch
- Cost Mismatch

Identify major operational risks.

6 Staff Productivity Analysis

Which warehouse staff members complete the highest number of operational tasks?

Compare:

- Completed tasks
 - Delayed tasks
 - Average completion time
 - Task efficiency
-

7 Delayed Task Analysis

Which task types are delayed most frequently?

- Picking
- Packing
- Receiving
- Dispatch
- Putaway
- Quality Check

Identify operational bottlenecks.

8 Discrepancy vs Task Delay Correlation

Analyze whether warehouses with:

- High delayed tasks also experience:
- High inventory discrepancies
- More validation issues

Create operational risk indicators.

Mail 4 – Executive Supply Chain Dashboard

👋 Dear Power BI Development Team,

After reviewing the warehouse operational dashboards, executive leadership now requires a centralized Executive Supply Chain Analytics Dashboard for strategic decision-making across all US distribution centers.

The COO and Supply Chain Director are facing major business concerns related to:

- Warehouse operational inefficiencies
- Overstock inventory
- Slow-moving stock
- High discrepancy warehouses
- Inventory validation failures
- Poor fill rate performance
- Inventory imbalance across warehouses
- Supplier-related operational issues

Leadership wants a high-level executive dashboard capable of monitoring overall supply chain health, operational performance, inventory risks, and warehouse benchmarking.

For this phase, the dashboard must focus heavily on:

- Advanced Data Modeling
- Executive KPI Benchmarking
- Advanced DAX Logic
- Dynamic Reporting
- Drill-through & Tooltip Pages
- Multi-page Dashboard Design
- Performance Benchmarking
- Operational Risk Indicators

The goal is to help executive leadership identify underperforming operational areas and improve overall inventory optimization strategy using Power BI.

Tables / Views Required

Tables

- item_categories
- suppliers
- warehouses
- warehouse_staff
- inventory_item_master
- inventory_movement_log
- inventory_profiling
- inventory_discrepancy_log
- inventory_validation_log
- task_allocation_log
- inventory_kpi_summary

Views

- vw_item_master_detail
- vw_inventory_movement_summary
- vw_inventory_profile
- vw_discrepancy_analysis
- vw_validation_issue_summary
- vw_task_productivity_summary
- vw_inventory_kpi_summary

Business Questions / Tasks

Executive Supply Chain Summary

Create an executive summary dashboard showing:

- Total stock value
 - Total inventory units
 - Total warehouses
 - Total discrepancy count
 - Total validation issues
 - Overall fill rate
 - Inventory turnover KPI
-

2 Warehouse Performance Benchmarking

Which warehouses are operationally underperforming?

Compare:

- Fill rate
- Inventory turnover
- Discrepancy count
- Validation issues
- Delayed tasks

Create warehouse performance rankings.

3 High Stock Value but Low Movement Analysis

Identify categories and warehouses where:

- Inventory stock value is high BUT
- Inventory movement is low

Detect potential dead stock and operational inefficiencies.

4 Discrepancy + Validation Risk Analysis

Which warehouses simultaneously have:

- High discrepancy value
- High validation issues
- High delayed task percentage

Create operational risk indicators.

5 Supplier Risk Analysis

Which suppliers are associated with:

- High discrepancy inventory items
- Frequent validation failures
- Slow-moving inventory

Identify supplier-related operational risks.

6 Fill Rate & Inventory Optimization Analysis

Which warehouses have poor fill rate performance?

Analyze:

- Overstock situations
- Stockout frequency
- Inventory turnover
- Slow-moving inventory

Recommend optimization focus areas.

7 Overstock & Slow-Moving Inventory Analysis

Which categories contain:

- High overstock inventory
- Slow-moving stock
- Low inventory turnover

Identify categories needing inventory optimization.

8 Executive Attention Dashboard

Create a dashboard highlighting:

- Critical warehouses
- High-risk inventory categories
- Operational bottlenecks
- Inventory imbalance
- Validation issue hotspots

This dashboard should help executives prioritize operational improvements.